RS exercise / linear system of equations

$$p(x,y) = f_0 y + f_1 xy + g_0 + g_1 x + g_2 x^2 + g_3 x^3 + g_4 x^4$$

plug-in pairs (x,y):

$(1,4)$, $(2,2)$, $(3,1)$, $(4,7)$, $(5,4)$, $(6,0)$

$$\begin{array}{ll}
\text{I} & 4f_0 + 4f_1 + g_0 + g_1 + g_2 + g_3 + g_4 = 0 \\
\text{II} & 2f_0 + 4f_1 + g_0 + 2g_1 + 4g_2 + 8g_3 + 16g_4 = 0 \\
\text{III} & f_0 + 3f_1 + g_0 + 3g_1 + 9g_2 + 27g_3 + 81g_4 = 0 \\
\text{IV} & 7f_0 + 28f_1 + g_0 + 4g_1 + 16g_2 + 64g_3 + 256g_4 = 0 \\
\text{V} & 4f_0 + 20f_1 + g_0 + 5g_1 + 25g_2 + 125g_3 + 625g_4 = 0 \\
\text{VI} & g_0 + 6g_1 + 36g_2 + 216g_3 + 1296g_4 = 0
\end{array}$$

→ reduce mod 11, write down coefficients → next page

Multiplicative inverses mod 11:

1	1	6	2
2	6	7	8
3	4	8	7
4	3	9	5
5	9	10	10

Gaussian Elimination

Pivot element

multiply 1st row by these factors to obtain zeros in 1st column (reduced mod 11)

	f_0	f_1	g_0	g_1	g_2	g_3	g_4	
I	4	4	1	1	1	1	1	
II	2	4	1	2	4	8	5	$\mid\cdot\left(-\frac{1}{2}\right) = -6 = 5$
III	1	3	1	3	9	5	4	$\mid\cdot\left(-\frac{1}{4}\right) = -3 = 8$
IV	7	6	1	4	5	9	3	$\mid\cdot\left(-\frac{7}{4}\right) = -7\cdot 3 = 1$
V	4	9	1	5	3	4	9	$\mid\cdot(-1) = 10$
VI	0	0	1	6	3	7	9	

multiplicators for 2nd row

I	4	4	1	1	1	1	1	
II	0	2	6	7	9	2	10	
III	0	2	9	0	6	2	1	$\mid\cdot(-1) = 10$
IV	0	10	2	5	6	10	4	$\mid\cdot\left(-\frac{10}{2}\right) = -5 = 6$
V	0	5	0	4	2	3	8	$\mid\cdot\left(-\frac{5}{2}\right) = -5\cdot 6 = 3$
VI	0	0	1	6	3	7	9	

III	0	0	3	4	8	0	2	
IV	0	0	5	3	5	0	9	$/ \cdot \left(-\frac{5}{3}\right) = -5 \cdot 4 = 2$
V	0	0	7	3	7	9	5	$/ \cdot \left(-\frac{7}{3}\right) = -7 \cdot 4 = 5$
VI	0	0	1	6	3	7	9	$/ \cdot \left(-\frac{1}{3}\right) = -4 = 7$

IV	0	0	0	0	10	0	2	pivot = 0
V	0	0	0	1	3	9	4	→ swap IV & V
VI	0	0	0	1	4	7	1	

V	0	0	0	1	3	9	4	
IV	0	0	0	0	10	0	2	✓
VI	0	0	0	1	4	7	1	$/ \cdot (-1) = 10$

IV	0	0	0	0	10	0	2	
VI	0	0	0	0	1	9	8	$/ \cdot \left(-\frac{1}{10}\right) = -10 = 1$

VI	0	0	0	0	0	9	10

complete	f_0	f_1	g_0	g_1	g_2	g_3	g_4
I	4	4	1	1	1	1	1
II	0	2	6	7	9	2	10
III	0	0	3	4	8	0	2
V	0	0	0	1	3	9	4
IV	0	0	0	0	10	0	2
VI	0	0	0	0	0	9	10

solve & plug-in bottom to top

Choose g_3 or g_4 arbitrarily $\neq 0$, e.g., $g_4 = 1$:

VI $\quad 9g_3 = -10 \longrightarrow g_3 = 5 \cdot 1 = 5$

IV $\quad 10g_2 + 2 = 0 \longrightarrow g_2 = -2 \cdot 10 = -20 = 2$

V $\quad g_1 + 3 \cdot 2 + 9 \cdot 5 + 4 = 0 \longrightarrow g_1 = -55 = 0$

III $\quad 3g_0 + 8 \cdot 2 + 2 = 0 \longrightarrow g_0 = -7 \cdot 4 = -28 = 5$

II $\quad 2f_1 + 6 \cdot 5 + 9 \cdot 2 + 2 \cdot 5 + 10 = 0$

$\quad 2f_1 = -35 = 9 \longrightarrow f_1 = 9 \cdot 6 = 54 = 10$

I $\quad 4f_0 + 4 \cdot 10 + 5 + 2 + 5 + 1 = 0$

$\quad 4f_0 = -53 = 2 \longrightarrow f_0 = 2 \cdot 3 = 6$

$$f(x) = 6 + 10x$$
$$g(x) = 5 + 2x^2 + 5x^3 + x^4$$

Final Steps :

$$-\frac{g(x)}{f(x)}$$